Percent Composition

Recall: Law of Constant Composition
Compounds always have the same percentage composition by mass.

The percent composition of a compound is a relative measure of the mass of each different element present in the compound.

Ex) 34.5 g of sodium combines with 53.3 g of chlorine to produce salt. What is the percent composition of sodium chlroride?

87.8g NaCl

$Cl = \frac{53.3}{87.8} = .607(100) = 60.7\%$

$Na = \frac{34.5}{87.8} = .393(100) = 39.3\%$

Ex) Sulfuric acid (H_2SO_4) consists of 0.6733 g of hydrogen, 10.69 g of sulfur and 21.33 g of oxygen. What is the percent composition of sulfuric acid?

$H_2SO_4 = 32.6933\ g$
$= 32.69$

$\%H = \frac{0.6733\ g}{32.69\ g} \times 100\%$
$= 2.060\%$

$S = \frac{10.69}{32.69} \times 100$
$= 32.70\%$

$O = \frac{21.33\ g}{32.69\ g} \times 100\%$
$= 65.25\%$

What is the % comp of $CaCl_2$?

Assume 1 mol

$Ca = 1\ mol \times 40.08\ g/mol$
$= 40.08\ g$

$Cl = 2\ mol \times 35.45\ g/mol$
$= 70.90\ g$

$Total = 110.98\ g$

Percent Composition worksheet

1. 52.9% Al
   47.1% O
2. 87.0% Ag
   13% S
3. 27.3% C
   72.7% O
4. 20.2% Al
   79.8% Cl
5. 32.30% C
   4.00% H
   63.70% O
6. 671 kg Zn